

Machine Learning

Machine Learning is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

There are three major types of Machine Learning: supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning uses labeled data to train a model. Common supervised learning algorithms include Linear Regression, Logistic Regression, Decision Trees, and K-Nearest Neighbors.

Unsupervised learning works with unlabeled data. Clustering is a common unsupervised learning technique used to group similar data points.

Reinforcement learning is based on an agent interacting with an environment. The agent receives rewards or penalties and learns to make better decisions.

Machine Learning is commonly used in recommendation systems, fraud detection, image recognition, healthcare, finance, and natural language processing.